

Topic 31 -- First Order Linear Equations

Definition of Linear

In a linear equation, the dependent variable and its derivatives are of first degree (i.e. raised to the first power). Any coefficients depend only on the independent variable.

Example 1

Is it linear?

$$y'' + 3xy' + 17y = 0$$

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$$3xy' + \sin y = 0$$

$$y'' + y' + \frac{17}{y} = 0$$

$$y' + 17y = e^x$$

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Definition of First-Order Linear

A first-order linear equation is an equation that may be written in the form:

$$\frac{dy}{dx} + P(x)y = Q(x) \text{ or } y' + P(x)y = Q(x)$$

Example 2

Solve: $y' + \frac{3}{x}y = x^4$

Example 3

Solve: $t \frac{dy}{dy} + 2t^2 y = t^2$

Example 4

Solve: $2xy' + y = 6x, x > 0, y(4) = 20$

Example 5

A tank contains 100 gal of brine made by dissolving 60 lb of salt in water. Salt water containing 1 lb of salt per gallon runs in at the rate of 2 gal/min and the well-stirred mixture runs out at the rate of 3 gal/min. Find the amount of salt in the tank after 30 minutes.